Gaussian distribution is arguably the most important single distribution (due to CLT) and it has a lot of interesting properties.

Univariate Gaussian distributions

$$f(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi}\sigma}\exp\left(-(x-\mu)^2/(2\sigma^2)\right)$$

Showing it is a proper distribution

$$\int_R f(x;\mu,\sigma)\,dx \cdot \int_R f(y;\mu,\sigma)\,dy$$

instead of solving $\int_R f(x;\mu,\sigma)\,dx$ directly, we use this multiplication trick here to simplify the problem

$$= \frac{1}{2\pi\sigma^2}\int_R\int_R \exp\left(-\frac{(x-\mu)^2}{2\sigma^2} - \frac{(y-\mu)^2}{2\sigma^2}\right)dx\,dy$$

$$= \frac{1}{2\pi\sigma^2}\int_R\int_R \exp\left(-\frac{1}{2\sigma^2}\left((x-\mu)^2+(y-\mu)^2\right)\right)dx\,dy$$

Let $(x-\mu) = r\cos\theta$, $(y-\mu) = r\sin\theta$

$$\Rightarrow J = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

$$\Rightarrow |J| = r\cos^2\theta + r\sin^2\theta$$

$$= r$$

Hence we have

$$\int_R f(x;\mu,\sigma)\,dx \int_R f(y;\mu,\sigma)\,dy$$

$$= \frac{1}{2\pi\sigma^2}\int_0^{2\pi}\int_0^{\infty} \exp\left(-\frac{1}{2\sigma^2}r^2\right) r\,dr\,d\theta$$

$$= \frac{1}{2\pi\sigma^2}\int_0^{2\pi}\int_0^{\infty} \frac{1}{2}\exp\left(-\frac{1}{2\sigma^2}r^2\right)dr^2\,d\theta$$

$$= \frac{1}{4\pi\sigma^2}\int_0^{2\pi}\left(-2\sigma^2\exp\left(-\frac{1}{2\sigma^2}r^2\right)\right)\Big|_{r=0}^{\infty}\,d\theta$$

$$= \frac{1}{4\pi\sigma^2}\int_0^{2\pi} 2\sigma^2\,d\theta$$

$$= \frac{1}{4\pi\sigma^2}\cdot 4\pi\sigma^2 = 1$$

derive its moment generating function

$$E(e^{Xt}) = \int_{-\infty}^{+\infty} e^{xt} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{xt - \frac{1}{2\sigma^2}(x^2+\mu^2-2\mu x)} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2}(x^2 - 2\mu x - 2\sigma^2 t x + \mu^2)} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2}((x-(\mu+\sigma^2 t))^2 - \sigma^4 t^2 - 2\mu\sigma^2 t)} dx$$

$$= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{\frac{1}{2}\sigma^2 t^2 + \mu t} \cdot e^{-\frac{1}{2\sigma^2}(x-(\mu+\sigma^2 t))^2} dx$$

$$= e^{\frac{1}{2}\sigma^2 t^2 + \mu t} \cdot \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\frac{1}{2\sigma^2}(x-(\mu+\sigma^2 t))^2} dx$$

$$= e^{\frac{1}{2}\sigma^2 t^2 + \mu t}$$

if $\mu = 0$, $\sigma = 1$, we have

$$E(e^{Xt}) = e^{\frac{1}{2}t^2}$$

$$\Rightarrow E(X) = \left(e^{\frac{1}{2}t^2} t\right)\Big|_{t=0} = 0$$

$$E(X^2) = \left(e^{\frac{1}{2}t^2} + t^2 e^{\frac{1}{2}t^2}\right)\Big|_{t=0} = 1$$

$$E(X^3) = \left(te^{\frac{1}{2}t^2} + 2te^{\frac{1}{2}t^2} + t^3 e^{\frac{1}{2}t^2}\right)\Big|_{t=0} = 0$$

$$E(X^4) = \left(e^{\frac{1}{2}t^2} + t^2 e^{\frac{1}{2}t^2} + 2e^{\frac{1}{2}t^2} + 2t^2 e^{\frac{1}{2}t^2} + 3t^2 e^{\frac{1}{2}t^2} + t^4 e^{\frac{1}{2}t^2}\right)\Big|_{t=0} = 3$$

Multivariate Gaussian distributions

$$f(x; \mu, \Sigma) = (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right)$$

proving it is a proper distribution:

$$I = \int_{R^n} \exp\left(-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1} (x-\mu)\right) dx$$

$$= \int_{R^n} \exp\left(-\tfrac{1}{2} y^T \Sigma^{-1} y\right) dy$$

if $z \sim N(0, I)$
then $\Sigma^{\frac{1}{2}} z + \mu \sim N(\mu, \Sigma)$

on the other hand, if
$z \sim N(\mu, \Sigma)$
$\Sigma^{-\frac{1}{2}}(z-\mu) \sim N(0, I)$

Since Σ is a valid covariance matrix, we must have

$\Sigma = S \Lambda S^T$, where S is orthonormal matrix, Σ is a diagonal matrix with $\Lambda_{i,i} = \lambda_i > 0$

$\Rightarrow \Sigma^{-1} = S \Lambda^{-1} S^T$

$\Rightarrow y^T \Sigma y = y^T S \Lambda^{-1} S^T y$

$= z^T \Lambda^{-1} z$, where $z = S^T y$

Note $\frac{\partial y}{\partial z} = \frac{\partial (Sz)}{\partial z} = S \Rightarrow |\det(S)| = 1$

because S is orthogonal matrix, we have
$SS^T = I$
$\Rightarrow |S||S^T| = 1$
$\Rightarrow \det(S) = \pm 1$

So we have

$$I = \int_{R^n} \exp\left(-\tfrac{1}{2} z^T \Lambda^{-1} z\right) dz$$

$$= \int_{R^n} \exp\left(\sum_{i=1}^n \tfrac{1}{\lambda_i} z_i^2\right) dz_1 \dots z_n$$

(λ_i: eigenvalues)

$$= \prod_{i=1}^n \sqrt{2\pi\lambda_i}$$

$$= (2\pi)^{n/2} |\Sigma|^{\frac{1}{2}}$$

$|\Sigma| = |S||\Lambda||S^T| = \prod_{i=1}^n \lambda_i$

mean and variance of the multivariate Gaussian

$$E(x) = \int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \, x \exp\left(-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) dx$$

$$= \int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} (y+\mu) \exp\left(-\tfrac{1}{2} y^T \Sigma^{-1} y\right) dy$$

$$= \mu + \underbrace{\int_{\mathbb{R}^n} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} y \exp\left(-\tfrac{1}{2} y^T \Sigma^{-1} y\right) dy}_{= 0 \text{ due to symmetry}}$$

$$= \mu$$

$$\mathrm{Cov}(X) = E\left((X-\mu)(X-\mu)^T\right)$$

$$= \int_{\mathbb{R}^n} (x-\mu)(x-\mu)^T (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\tfrac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)\right) dx$$

$$= \int_{\mathbb{R}^n} y y^T (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\tfrac{1}{2} y^T \Sigma^{-1} y\right) dy$$

Let $z = \Sigma^{-\frac{1}{2}} y$, $\Rightarrow y = \Sigma^{\frac{1}{2}} z$

$$\Rightarrow = \int_{\mathbb{R}^n} \Sigma^{\frac{1}{2}} z z^T \Sigma^{\frac{1}{2}} (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\tfrac{1}{2} z^T z\right) \left|\det\left(\Sigma^{\frac{1}{2}}\right)\right| dz$$

$$= \Sigma^{\frac{1}{2}} \underbrace{\int_R z z^T (2\pi)^{-\frac{n}{2}} \exp\left(-\tfrac{1}{2} z^T z\right) dz}_{\text{all independent now}} \Sigma^{\frac{1}{2}}$$

$$= \Sigma$$

Conditional distribution of multivariate Gaussian

If $(X, Y) \sim \mathcal{N}\left(\begin{pmatrix} \mu_x \\ \mu_y \end{pmatrix}, \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{xy} & \Sigma_{yy} \end{pmatrix} \right)$, then $Y|X \sim$?

μ $\quad$ Σ

$$f_{X,Y}(x, y; \mu, \Sigma) = (2\pi)^{-\frac{n}{2}} \det(\Sigma)^{-\frac{1}{2}} \exp\left(-\frac{1}{2} \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix} \right)$$

Let $\Lambda = \Sigma^{-1}$ (precision matrix)

$$= (SAS^T)^{-1}$$

$$= S A^{-1} S^T \quad \text{→ it is positive-definite as well}$$

$$= \begin{pmatrix} \Lambda_{xx} & \Lambda_{xy} \\ \Lambda_{yx} & \Lambda_{yy} \end{pmatrix}, \text{ where } \Lambda_{xx} \text{ and } \Lambda_{yy} \text{ are symmetric}$$

$\Lambda_{xy} = \Lambda_{yx}^T$

$$\Rightarrow -\frac{1}{2} \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix} = -\frac{1}{2} \Big((x-\mu_x)^T \Lambda_{xx} (x-\mu_x) + (y-\mu_y)^T \Lambda_{yx} (x-\mu_x)$$

$$+ (x-\mu_x)^T \Lambda_{xy} (y-\mu_y) + (y-\mu_y)^T \Lambda_{yy} (y-\mu_y) \Big)$$

Note that the conditional function $f_{Y|X}(y|x; \mu, \Sigma) = \dfrac{f_{X,Y}(x, y; \mu, \Sigma)}{f_X(x; \mu, \Sigma)}$ is a function of y

and $-\frac{1}{2} \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix}^T \Sigma^{-1} \begin{pmatrix} x-\mu_x \\ y-\mu_y \end{pmatrix} = -\frac{1}{2} \Big(y^T \Lambda_{yy} y - y^T \Lambda_{yy} \mu_y - \mu_y^T \Lambda_{yy} y + y^T \Lambda_{yx} (x-\mu_x)$

$$+ (x-\mu_x)^T \Lambda_{xy} y + \text{Const} \Big)$$

$$= -\frac{1}{2} \Big(y^T \Lambda_{yy} y + y^T \big(2\Lambda_{yx}(x-\mu_x) - 2\Lambda_{yy}\mu_y \big) \Big) + \text{const}$$

$$= -\frac{1}{2} \Big(y^T \Lambda_{yy} y + 2 y^T \Lambda_{yy} \big(\Lambda_{yy}^{-1} \Lambda_{yx} (x-\mu_x) - \mu_y \big) \Big) + \text{const}$$

Note that any multivariate Gaussian distribution has the form

$$(2\pi)^{-n/2} \det(\Sigma)^{-\frac{1}{2}} \exp\left\{ -\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu) \right\}$$

$$\rightarrow -\frac{1}{2} (x-\mu)^T \Sigma^{-1} (x-\mu)$$

$$= -\frac{1}{2} \left(x^T \Sigma^{-1} x - x^T \Sigma^{-1} \mu - \mu^T \Sigma^{-1} x + \mu^T \Sigma^{-1} \mu \right)$$

$$= -\frac{1}{2} \left(x^T \Sigma^{-1} x - 2 x^T (\Sigma^{-1}) \mu \right) + \text{Constant}$$

$x^T \Sigma^{-1} x$: quadratic term $\quad$ $x^T (\Sigma^{-1}) \mu$: linear term

Therefore, we must have the covariance of the conditional function

$$Cov(Y|X) = \Lambda_{yy}^{-1}$$

$$E(Y|X) = \Lambda_{yy}^{-1}\Lambda_{yx}(X-\mu_x) - \mu_y$$

Symmetric $\Leftrightarrow$

$$Cov(X|Y) = \Lambda_{xx}^{-1}$$

$$E(X|Y) = \Lambda_{xx}^{-1}\Lambda_{xy}(y-\mu_y)-\mu_x$$

Now, the problem is to express the precision matrix

$$\Sigma^{-1} = \Lambda = \begin{pmatrix} \Lambda_{xx} & \Lambda_{xy} \\ \Lambda_{yx} & \Lambda_{yy} \end{pmatrix} \quad \text{using} \quad \Sigma = \begin{pmatrix} \Sigma_{xx} & \Sigma_{xy} \\ \Sigma_{yx} & \Sigma_{yy} \end{pmatrix}$$

To do this, we apply Schur complement based on Gaussian elimination

$$\begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix} = \begin{pmatrix} A-BD^{-1}C & B \\ 0 & D \end{pmatrix}$$

$$\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}\begin{pmatrix} A-BD^{-1}C & B \\ 0 & D \end{pmatrix} = \begin{pmatrix} A-BD^{-1}C & 0 \\ 0 & D \end{pmatrix}$$

So we have

$$\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}\begin{pmatrix} A & B \\ C & D \end{pmatrix}\begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix} = \begin{pmatrix} A-BD^{-1}C & 0 \\ 0 & D \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} A & B \\ C & D \end{pmatrix} = \begin{pmatrix} I & BD^{-1} \\ 0 & I \end{pmatrix}\begin{pmatrix} A-BD^{-1}C & 0 \\ 0 & D \end{pmatrix}\begin{pmatrix} I & 0 \\ D^{-1}C & I \end{pmatrix}$$

$$\Rightarrow \quad \begin{pmatrix} A & B \\ C & D \end{pmatrix}^{-1} = \begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix}\begin{pmatrix} (A-BD^{-1}C)^{-1} & 0 \\ 0 & D^{-1} \end{pmatrix}\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} I & 0 \\ -D^{-1}C & I \end{pmatrix}\begin{pmatrix} M & 0 \\ 0 & D^{-1} \end{pmatrix}\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} M & 0 \\ -D^{-1}CM & D^{-1} \end{pmatrix}\begin{pmatrix} I & -BD^{-1} \\ 0 & I \end{pmatrix}$$

$$= \begin{pmatrix} M & -MBD^{-1} \\ -D^{-1}CM & D^{-1}CMBD^{-1}+D^{-1} \end{pmatrix}$$

where $M = (A-BD^{-1}C)^{-1}$

Generate non-identity Gaussian distribution.

Suppose we can sample from $N(0,1)$. How to generate $X \sim N(\mu, \Sigma)$?

Let $Y \sim N(0,1)$, and $Z = \mu + WY$, then

$$E(Z) = \mu$$
$$Cov(Z) = W\, Cov(Y) W^T = WW^T$$

$\Rightarrow$ if $WW^T = \Sigma$, then $Z \sim N(\mu, \Sigma)$

1. Cholesky decomposition. For a real-valued symmetric positive definite matrix A, we have

$$A = LL^T$$

where L is a lower triangular matrix with strictly positive diagonal entries.

2. Eigen-decomposition. Since A is symmetric, we have

$$\begin{aligned} A &= Q\Lambda Q^T \\ &= Q\Lambda^{\frac{1}{2}}\Lambda^{\frac{1}{2}}Q^T \\ &= Q\Lambda^{\frac{1}{2}}(Q\Lambda^{\frac{1}{2}})^T \\ &= WW^T \end{aligned}$$

3. Let $A^{\frac{1}{2}} = Q\Lambda^{\frac{1}{2}}Q^T \Rightarrow A = A^{\frac{1}{2}}A^{\frac{1}{2}} = A^{\frac{1}{2}}(A^{\frac{1}{2}})^T$

↓ this W is symmetric

If W works, i.e. $WW^T = \Sigma \Rightarrow WR$ also works if $RR^T = I$

Since $WRR^TW^T = WW^T = \Sigma$

We have infinitely many valid matrix W.